HireRL: Multi-Agent Reinforcement Learning as an Alternative to Stable Matching under Informational Frictions

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Abstract

Stable matching is a foundational algorithmic framework for organizing two-sided allocation problems, with applications ranging from school choice and residency assignment to online platforms. However, most canonical formulations assume that match quality is fixed and fully observable, abstracting away from realistic settings in which agents must acquire information through costly and noisy interactions. In many markets, preferences and match outcomes are only partially revealed over time, creating trade-offs between screening effort, mismatch risk, and separation costs. We investigate whether multi-agent reinforcement learning can provide a simulation tool to study two-sided matching under endogenous information acquisition and partial observability. We develop **HireRL**, a benchmark that reflects a competitive labor market grounded in the labor economics literature. It incorporates essential components such as screening, wage determination, matching, profit generation, and continuation decisions. Our experiments show that, under information frictions in the hiring process, learned decentralized policies can outperform static stable-matching baselines in terms of aggregate social welfare. The core contribution of HireRL is a calibratable MARL simulation environment that enables the study of dynamic strategic behavior in matching mechanisms beyond the classical stable matching mechanism. More broadly, HireRL provides a general tool for investigating how learning and information frictions reshape equilibrium outcomes in two-sided matching problems.

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1. Introduction

Stable matching is a classical algorithmic framework for organizing two-sided allocation problems, in which two sets of agents must be paired according to preferences or utilities. Since the seminal work of Gale & Shapley (1962), stable matching mechanisms have served as a canonical solution concept for markets ranging from school choice and medical residency assignment (Abdulkadiroğlu & Sönmez, 2003; Roth, 1996) to online platforms and resource allocation in distributed systems (Jorswieck, 2011; Hasan & Hossain, 2015). These mechanisms produce matching with no blocking pairs, ensuring a form of equilibrium stability under centralized allocation.

Despite their broad applicability, most stable matching formulations rely on strong informational assumptions: match quality is typically treated as fixed, fully observed, and known to agents at the time of matching. In contrast, many real-world matching environments (Nguyen & Sila, 2025) involve substantial uncertainty, where agents must strategically acquire information through costly screening, interaction, or experimentation before forming long-term matches. This introduces dynamic trade-offs between exploration effort, mismatch risk, and the cost of reversing poor assignments.

Such informational frictions naturally motivate a more realistic modeling of the matching problem. There is a nascent body of work using multi-agent reinforcement learning (MARL) to study real-world problems like tax policy design (Zheng et al., 2020) and climate investment (Hou et al., 2025). While these works have shown promise on studying real-world problems using MARL, they have not yet directly compared a MARL solution to a solution from canonical algorithmic game theory. In this work, we study the two-sided market matching problem using MARL and compare it to the stable matching algorithm. Rather than treating matching as a centralized optimization problem, MARL enables the simulation of decentralized agents who learn policies under partial observability, strategic competition, and endogenous belief updating. This provides a richer modeling framework for studying matching mechanisms when preferences and payoffs emerge through interaction over time, rather than being fully specified upfront.

To enable systematic study of matching under dynamic information acquisition, we propose a multi-agent reinforcement learning benchmark that captures decentralized matching behavior and heterogeneous agent modeling in the presence of screening costs, noisy signals, and separation decisions. We use the labor market as the matching environment, as it is extensively studied (Shi, 2025) and it is a real-world problem of practical and economic significance. While our environment is motivated by hiring as a concrete instantiation, the underlying framework applies more broadly to two-sided markets in which match values are partially observable and must be learned over time. Compared to stable matching, we show that MARL solutions achieve higher aggregate welfare on noisy, dynamic two-sided allocation environments. Furthermore, the benchmark supports flexible control of the environment parameters, enabling counterfactual evaluation of how informational frictions reshape stability and welfare relative to classical stable matching baselines.

Our contributions are summarized as follows:

- We introduce **HireRL**, a simulation platform and MARL benchmark for two-sided matching markets. It enables the study of decentralized screening, wage offers, match formation, and continuation decisions under realistic labor market frictions. We release HireRL as an open-source tool to support reproducible research and algorithm development in dynamic labor market settings.

- Our results (Section 5.1) demonstrate that the environment captures economically reasonable behaviors, including the negative relationship between tenure and turnover (Figure 9) and higher interview investment by richer firms (Figure 14). Building on this foundation, we provide quantitative evidence (Section 5.2) that learned decentralized policies can outperform static stable-matching baselines in terms of aggregate social welfare under noisy match evaluation, while inducing new forms of instability.

- We highlight multi-agent reinforcement learning as a powerful computational tool for studying economic mechanism design problems under realistic frictions, enabling the analysis of dynamic strategic behavior and equilibrium outcomes that are analytically intractable in large matching markets. We provide insight into why screening intensity differs across real-world labor markets: platforms like gig economy companies can rely on rapid performance feedback, whereas firms like high-tech companies face high uncertainty and costly mis-hires, necessitating substantial upfront interviewing. The empirical results will be presented in Section 5.4

2. Related Works

Stable matching. Stable matching provides a centralized allocation rule that eliminates *blocking pairs*—pairs of agents who would both prefer to be matched with each other over their assigned partners—and ensures that no (firm, worker) pair has an incentive to deviate from the outcome. Recent work extends this tradition to settings with incomplete or noisy information. For example, Rastegari et al. (2016) analyzes preference elicitation via interviews in stable marriage markets, while Taywade et al. (2021) studies decentralized learning dynamics in stable matching environments under noisy preferences, using MARL to recover stable and fair outcomes when rankings are incomplete or uncertain. In contrast, our goal is not to enforce stability as the primary objective or to learn stable matchings under partial rankings. Instead, we model screening as a costly and endogenous decision made by profit-maximizing firms, and study how decentralized incentives shape hiring, separation, and welfare when blocking-pair constraints are not imposed. Rather than treating interviews as a mechanism to restore stability, we focus on how agents dynamically trade off screening effort against hiring costs and mismatch risk.

Learning-based matching. A complementary strand shifts attention away from guaranteeing stability and toward learning and information acquisition in matching environments. Pandora's box models investigate costly inspection and optimal stopping under uncertainty. Recent extensions allow for combinatorial cost structures and multi-stage probing mechanisms, closely mirroring interviews of varying intensity (Beyhaghi & Cai, 2024; Berger et al., 2023; Hoefer et al., 2024). Related work on online matching with stochastic rewards and bandit learning emphasizes welfare, regret, and incentives rather than stability (Mehta & Panigrahi, 2012; Li et al., 2025). Within this line, Li et al. (2022) propose a dynamic matching bandit framework for two-sided online markets in which preferences are initially unknown and must be learned over time. Our setting differs in that screening decisions are made repeatedly by strategic firms in a labor market where worker productivity evolves over long horizons, and where interview effort directly affects both costs and future match quality.

Reinforcement learning for economics and matching platforms. A growing literature uses MARL as a simulation tool for studying economic mechanisms and institutional design problems, including tax policy and redistribution in the AI Economist framework (Zheng et al., 2020), LLM-based economic agent simulations (Karten et al., 2025), and climate investment dynamics in ESG-driven markets (Hou et al., 2025). These works highlight the promise of MARL for modeling decentralized strategic behavior under uncertainty in complex economic systems.

Moreover, Azagirre et al. (2024) documents the deployment of a reinforcement learning matching system at Lyft, optimizing long-run platform efficiency and earnings rather than stability. However, gig-economy platforms often operate in environments where match quality is revealed quickly through standardized tasks and immediate feedback. In traditional labor markets, by contrast, worker productivity is difficult to observe ex ante and is learned only gradually through sustained employment relationships. HireRL is designed specifically to capture this costly learning process, making it suitable for studying optimal screening policies in labor-market settings.

Employer learning and labor-market frictions. In this work, we focus on the labor market as a two-sided matching problem with noisy information and dynamic match formation, motivated by its broad relevance in real-world hiring settings. Labor markets have also been extensively studied in labor economics as settings where match quality is only partially observed and must be learned over time.

A large body of work on employer learning documents how firms gradually infer workers' productivity and how this process shapes wage dynamics and inequality (Farber & Gibbons, 1996; Lange, 2007; Schönberg, 2007; Lee & Schwarz, 2017). While this literature delivers equilibrium characterizations with transparent functional forms, optimal policies are derived analytically rather than learned through trial-and-error, as in reinforcement learning. Our work complements this tradition by introducing a flexible multi-agent reinforcement learning (MARL) simulation benchmark that accommodates a wide range of labor market environments. By allowing firms to endogenously choose their screening intensity, the platform facilitates counterfactual analysis of policy interventions—such as changes in firing costs—and their effects on unemployment, match stability, and welfare.

3. HireRL as A Two-Sided Matching Environment

We model the labor market as a dynamic matching market with Markovian dynamics. We introduce **HireRL**, a multi-agent reinforcement learning benchmark for two-sided matching under *noisy observation* and *sequential learning about worker ability*. HireRL includes strategic firms $\mathcal{C}$ (decision-making agents, controlled by MARL policies) and a pool of workers $\mathcal{W}$ (modeled as part of the environment). The objective of a firm is to maximize its reward (defined as the profit), and the worker accepts the offer with the maximum wage. While the HireRL environment does not encompass all matching settings, it serves as a realistic and tractable simplification that enables us to test key questions about how MARL-based compares to stable matching.

HireRL is built upon the following core concept: σ_j denotes the latent ability of worker j. Firms cannot directly observe σ_j; instead, firm i learns a private estimate $\tilde{\sigma}_{ij}$ by incurring an interview cost. A public label $\hat{\sigma}_j$ is observable to all firms by adding noise to σ_j. To capture dynamic agent evolution, at each step when worker j is employed by firm i, the worker's experience e_j accumulates over time, and the firm gradually updates its private estimate $\tilde{\sigma}_{ij}$. Figure 1 illustrates the overall process flow of **HireRL**.

3.1. MARL Formulation

We leverage MARL to model the decentralized decision-making process of self-interested firms. Classical mechanisms like stable matching assume a centralized planner and fully specified preferences. However, in reality, firms compete with each other and independently learn hiring strategies under partial observability, adapting to noisy information acquisition and evolving employment histories. In our experiments, we train heterogeneous firms to maximize their own rewards and evaluate the resulting matching behavior. We adopt Independent PPO (Schulman et al., 2017; De Witt et al., 2020), where each firm learns its own policy without parameter sharing or centralized coordination (Lowe et al., 2017; Rashid et al., 2020; Yu et al., 2022). We summarize the main component of **HireRL** below.

Observations. Firms do not observe worker ability directly. Instead, they rely on noisy public signals (e.g., resumes) and imperfect private information obtained through costly interviews. Over time, realized job performance provides additional information that refines firms' beliefs about worker quality.

Actions. At each round, each firm i chooses an individual screening investment level $c_{ij}^{\text{interview}}$ for each unemployed worker j in the market. Higher screening intensity yields more accurate evaluations but incurs higher interview costs, creating a trade-off between information acquisition and profitability.

Rewards and Separation. Firms are rewarded by the net surplus generated from their workforce. Specifically, firm i (with the current employee set E_i) receives per-period reward

$$r_i = \sum_{j \in E_i} \left(p_{ij} - w_{ij} - c_{ij}^{\text{interview}} - c_{ij}^{\text{fire}} \right),$$

where productivity p_{ij} depends on worker ability and experience, while wages w_{ij} are determined by the firm's evolving private beliefs. Interview and firing costs capture frictions in the matching and separation process (Sections 3.2 and 3.3). Separation occurs only through firm-initiated fir-

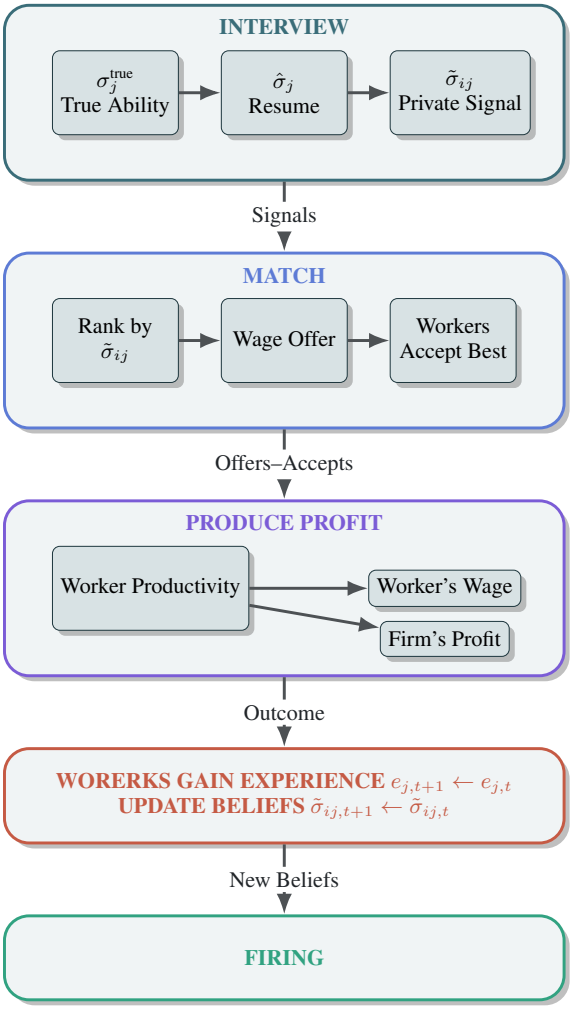

Figure 1. Overview of HireRL environment dynamics. The process proceeds through screening, matching, and production, culminating in belief updates, experience accumulation, and firing process.

ing: a worker j is dismissed by firm i whenever

$$p_{ij} - w_{ij} < c \cdot w_{ij},$$

with $c = 0.9$, meaning firms fire employees who generate less than 90% of their wage in net surplus.

Dynamics. The market evolves endogenously as workers gain experience on the job, firms update beliefs from observed outcomes, wages adjust over time, and low-surplus matches may dissolve. This produces a dynamic matching environment in which decentralized learning policies can be systematically compared against classical stable-matching benchmarks.

In the following sections, we detail the micro-founded design of **HireRL**, focusing on firms' information acquisition and learning, match profit realization, and wage updating mechanisms.

3.2. Matching by Incurring Interview Cost

This part of the environment models the noisy matching process. Firm i can invest a cost for the interview process to obtain a private ability estimate $\tilde{\sigma}_{ij}$ of worker j. Paying a higher interview cost yields a larger belief revision.

At the beginning of the episode, Workers are born with latent ability $\sigma_j \in [0, 1]$, drawn from a truncated Gaussian:

$$\sigma_j \sim \mathcal{N}(0.5, 1) \mid \sigma_j \in [0, 1]. \tag{1}$$

At the same time, all firms observe a noisy public label $\hat{\sigma}_j = \sigma + \epsilon_j$ (e.g., with a Gaussian noise), which represents coarse pre-hiring information such as résumé quality or standardized test performance. Firm i may then invest $c_{ij}^{\text{interview}}$ to obstain a more informative private signal:

$$\tilde{\sigma}_{ij} = \hat{\sigma}_j - \eta\left(c_{ij}^{\text{interview}}, \hat{\sigma}_j - \sigma_j\right) \operatorname{sgn}(\hat{\sigma}_j - \sigma_j).$$

The step size

$$\eta_{ij} = \frac{(\hat{\sigma}_{j,t} - \sigma_j)^2}{1 + (\hat{\sigma}_{j,t} - \sigma_j)^2} \cdot \frac{(\ell c_{ij}^{\text{interview}})^2}{1 + \ell c_{ij}^{\text{interview}}}$$

is monotone to the interview cost $C_{\text{interview}}$, reflecting the fact that more intensive interviews generate more informative signals. The environment parameter l is chosen so that $\ell C_{\text{interview}} \leq 2.7$ to ensure numerical stability (see App. A for proof).

After the interview process, firms issue the initial wage offers:

$$w_{ij,t}^{\text{offer}} = \phi_{\text{type}(i)}\, g\big(\tilde{\sigma}_{ij,t=\text{interview}}\big)\, I_{ij,t}^{\text{offer}}, \quad \text{where} \quad g = \tfrac{1}{2}\big(1 + \tanh(0.5\,\tilde{\sigma})\big) \tag{2}$$

to the top $\lfloor \kappa N_{\text{int}} \rfloor$ candidates ranked by the private estimate $\tilde{\sigma}$. And workers accept the highest-wage offer.

3.3. Agent Evolving Dynamics

While stable matching typically computes a fixed solution, real-world matching problems are often dynamic. We introduce two agent evolution mechanisms in **HireRL**: (1) experience increases future productivity, and (2) workers' latent ability is revealed gradually through interviewing(3.2) and employment histories. The experience updates as:

$$e_{j,t+1} = e_{j,t} + (g_0, g_1\sigma_j)\mathbf{1}\{j \text{ employed at } t\}e^{-\theta e_{j,t}},$$

with diminishing returns governed by $\theta > 0$. This captures learning-by-doing dynamics consistent with Altonji & Pierret (2001); Schönberg (2007): productivity rises with experience, but gains are largest early in a worker's career.

As discussed in Section 3.2, the discrepancy between perceived ability $\tilde{\sigma}$ and true ability σ is largest in a worker's

early career, and narrows over time as firms accumulate more information. The private belief is updated as a convex combination of the interview belief and a performance-based signal, which is formulated by

$$\tilde{\sigma}_{ij,t+1} = (1 - v_{j,t})\,\tilde{\sigma}_{ij,\text{interview}} + v_{j,t}\,\sigma_j,$$
$$v_{j,t} = \frac{\exp_{j,t} K_1}{1 + (\exp_{j,t} - 1)K_1}, \qquad K_1 = \frac{\delta^2_{\text{interview}}}{\delta^2_{\text{interview}} + \delta^2_{\epsilon}}. \tag{3}$$

This matches the core implication of employer learning: as tenure increases, firms rely less on initial signals and more on realized productivity (Farber & Gibbons, 1996; Altonji & Pierret, 2001; Schönberg, 2007; Lange, 2007).

$\hat{\sigma}_{j,t}$: the public signal (e.g., from a resume or LinkedIn) of a worker j in any timesteps. If worker j is not hired in period t, the public belief remains unchanged as $\hat{\sigma}_{j,t+1} = \hat{\sigma}_{j,t}$..

When the worker is hired at step $t_{\text{interview}}$, the public belief updates in the same direction as the private belief, but with a smaller adjustment:

$$\hat{\sigma}_{j,t+1} = \tilde{\sigma}_{j,t_{\text{interview}}} + \tfrac{1}{2}\Big(\tilde{\sigma}_{j,t+1} - \tilde{\sigma}_{j,t_{\text{interview}}}\Big).$$

$$\hat{\sigma}_{j,t+1} = \tfrac{1}{2}\,\tilde{\sigma}_{j,t_{\text{interview}}} + \tfrac{1}{2}\,\tilde{\sigma}_{j,t+1}.$$

As the worker's experience and the firm's private estimate evolves over time, the worker's productivity p_{ij} and wage w_{ij} will also be updated. When worker j is employed by firm i, the productivity is defined as:

$$p_{ij,t} = \text{COMPANY TYPE} \times \Big(f(\exp_{j,t-1} + (g_0 + g_1\sigma_j)\mathbf{1}\{j \text{ employed at } t-1\}e^{-\theta \exp_{j,t-1}}) + \epsilon_{ij,t}\Big),$$
$$\epsilon_{ij,t} \sim \mathcal{N}(0, \delta^2_{\epsilon}), \tag{4}$$

Here, COMPANY TYPE is a linear scalar multiplier, so some firms convert the same worker input into higher profit; consequently, firms are heterogeneous. $f(\cdot)$ is increasing with diminishing returns. Thus, expected output is increasing in ability, so on-the-job profit provides a noisy signal of worker quality, consistent with employer-learning models (Lange, 2007).

Wages are also dynamically adjusted to reflect both initial screening and realized profitability:

4. Experiment Setup

Environment configuration. Unless otherwise stated, we simulate a market with 5 HETEROGENEOUS FIRMS and 120 WORKERS. Each firm has a capacity of 20. We evaluate policies over 200 EPISODES, each of length 100 timesteps, with Gaussian noise in the worker's public labels ($\epsilon_j = \hat{\sigma}_j - \sigma_j \sim \mathcal{N}(0, 0.2^2)$). All results are averaged over at least five independent random seeds, and the error bars indicate the standard error with 95% confidence intervals.

4.1. Baselines

Random policy. We test HireRL against a random policy which mirrors our baseline policy but replaces firm decisions with stochastic behavior while keeping wage, profit, and reward mechanics intact.

Stable matching baseline. We consider a many-to-one matching market with firm capacities and unit-demand workers. At the start of each episode, worker types and firms' initial beliefs are drawn, inducing strict preference rankings. We compute outcomes using the firm-proposing Deferred Acceptance (DA) algorithm (Gale & Shapley, 1962).

Unlike **HireRL**, which is dynamic and allows firms to learn and update beliefs over repeated interactions, DA is static and one-shot: firms rank workers solely based on initial public information $\hat{\sigma}_j$ (i.e., resume-only screening with no interviews or learning). Accordingly, we evaluate DA in a single-step episode, whereas HireRL runs for 100 steps.

Implementation details. To ensure comparability, we align both environments at reset. Workers rank firms by offered wages, and firms rank workers by their initial belief $\tilde{\sigma}_{t=0}$, which remains fixed in DA. We export the reset-time draws from HireRL—including σ, $\hat{\sigma}$, $\tilde{\sigma}$, and noise parameters—to initialize the DA baseline with the same worker identities. Since experience does not evolve in a one-shot framework, we fix worker experience in DA to the realized end-of-episode experience from HireRL, matched by the common initial $\hat{\sigma}_j$ ordering. For example, if worker j starts with $\hat{\sigma}_j = 0.5$ and ends a HireRL episode with experience $e_j^{\text{end}} = 0.8$, we export $(\hat{\sigma}_j, e_j^{\text{end}})$ and assign the same experience value to the corresponding worker in the stable matching baseline (matched by the shared initialization of $\hat{\sigma}_j$).

To further match the dynamics of HireRL, firms in the DA baseline are also allowed to fire workers after the Gale–Shapley allocation using the same profitability rule as in Section 3.1. Note that DA guarantees stability (zero blocking pairs) for the pre-firing matching, but does not ensure full employment. Post-matching firing may increase unemployment, and we include it for a fair welfare comparison with HireRL.

4.2. Hypotheses

Our experiments investigate the following questions.

H1: ENVIRONMENT VALIDITY CHECK. Does HireRL generate qualitatively reasonable market statistics and behavioral regularities (e.g., tenure–turnover patterns, meaningful screening expenditure, persistent employment), consistent with economic literature and empirical facts?

H2: ALGORITHM CONTRIBUTION. Will **HireRL** outperform stable matching in terms of social welfare and firm reward as screening signals become noisier, and match quality becomes harder to infer?

H3: POLICY TOOL. Can **HireRL** be used to help answer policy-driven questions? In this work, we study the following question: does how higher unemployment benefits (modeled as higher firing costs) affect social welfare and firm profit?

H4: SCREENING HETEROGENEITY FROM DECENTRALIZED LEARNING. How does dispersion in true ability affect hiring costs? We hypothesize that when the applicant pool exhibits greater ability heterogeneity, firms have stronger incentives to invest in screening, making higher hiring expenditure optimal in order to reduce mismatch risk.

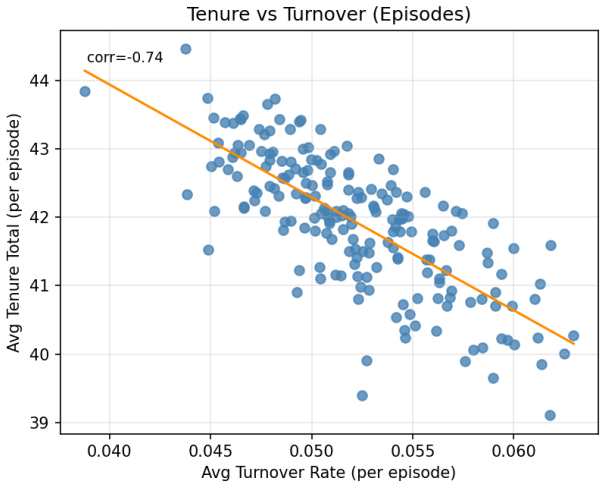

Figure 2. Turnover rate vs. cumulative tenure across evaluation episodes. Each point represents one episode (episode-mean turnover on the x-axis; episode-mean cumulative tenure on the y-axis). The orange line shows a least-squares fit.

5. Main results

5.1. H1: Environment validity check

Our results demonstrate that **HireRL** generates economically plausible labor-market statistics. For example, consistent with established theory and empirical evidence (Topel & Ward, 1992; Munasinghe, 2000), job tenure is negatively related to turnover: separation rates decline as workers accumulate tenure (Fig. 2).

Figure 3 further illustrates realistic firm-side behavior by reporting the interview cost-to-reward ratio across firms. Firms with higher company type (`ctype`, meaning they generate higher profit from the same set of workers) and that offer larger wage premiums, as commonly observed among large firms (Oi & Idson, 1999), incur systematically greater screening expenditures relative to rewards. This reflects the higher effort required to identify suitable candidates for more demanding and higher-paying positions.

Figure 8 also shows that increasing firing costs—which can be interpreted as a lump-sum deadweight loss in our environment—reduces aggregate social welfare. This is because firing costs are not modeled as transfers to workers (e.g., unemployment benefits), but instead represent pure waste. This pattern is consistent with basic economic intuitions.

Together, these patterns provide evidence that **HireRL** provides a valid and economically grounded environment for studying two-sided matching problems.

5.2. H2: Algorithm contribution.

HireRL increases social welfare and becomes more valuable as noise increases. Figures 4 & 5 show that HireRL consistently achieves higher social welfare than the alternative policies. The performance advantage of HireRL persists under different interview noise regimes. One possible intuition lies in the contrast between centralized allocation and decentralized market forces, consistent with the literature showing that stability can come at a welfare cost—the "price of stability" (Boudreau & Knoblauch, 2014). Stable matching resembles a planner-imposed mechanism that enforces stability constraints, whereas HireRL allows agents to adapt strategically through learning, yielding more efficient outcomes in noisy and dynamic environments. In this sense, HireRL captures how decentralized decision-making can sometimes improve aggregate welfare when preferences and match quality are not fully known ex ante.

Figure 7 reports per-firm rewards across noise levels. The relative gains of **HireRL** are larger when signals are noisier: as screening becomes harder and search frictions intensify, static baselines cannot adapt, while **HireRL** adjusts screening intensity and hiring decisions endogenously. **HireRL** reduces screening waste and raises firm profits compared with random policy, at the cost of additional instability as shown in Figure 6. Notably, stable matching yields the highest outcomes for firms 0 through 3. This is consistent with economic intuition and literature (Campos et al., 2025): under a benevolent, centralized, planner-designed mechanism, allocations can be optimized to improve overall efficiency, whereas in a competitive market with self-interested agents, gains are typically unevenly distributed—some firms benefit

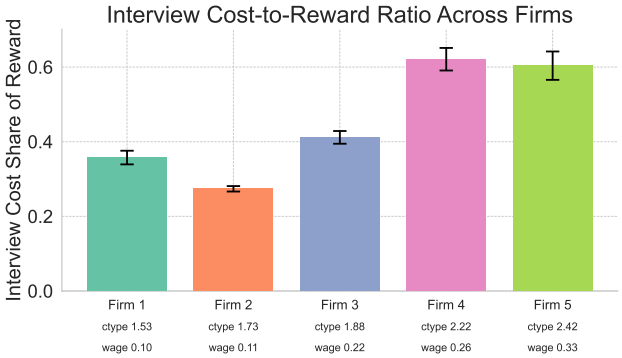

Figure 3. Interview cost-to-reward ratio across firms under HireRL. Bars report the mean ratio of interview cost per step to step-average reward, with error bars showing 95% confidence intervals across seeds. Firms with higher company type (`ctype`, meaning companies that receive a higher profit premium) and wages incur larger screening costs relative to rewards, reflecting increased effort required to identify suitable workers.

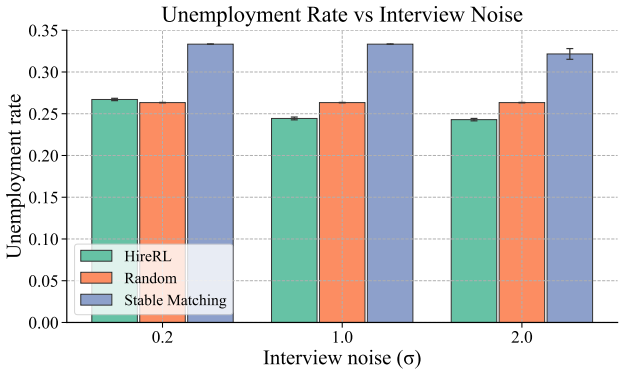

Figure 4. Unemployment rate under varying levels of interview noise for HireRL, a random matching baseline, and stable matching. Unemployment rate $=$ Unemployed/Labor force. Under HireRL, unemployment slightly decreases as noise increases, suggesting that learned policies adapt better compared with others. Across all noise levels, HireRL maintains consistently the lowest unemployment, highlighting the advantage of decentralized learning under imperfect information. The error bars are 95% confidence intervals across seeds.

while others may be worse off relative to the centralized benchmark.

5.3. H3: Policy tool

HireRL can also be used as a policy evaluation tool by varying key labor-market frictions. When firing costs increase (interpretable as stronger employment protection or more generous unemployment benefits), firms face higher separation costs and must account for the greater expense of terminating unproductive matches. This changes firms' incentives ex ante, making them more cautious and selective in hiring decisions, which can reduce overall match formation.

Figures 8–10 illustrate the impact of raising firing costs under different levels of signal noise. Higher firing costs consistently reduce aggregate welfare (Figure 8), reflecting increased labor-market frictions and less efficient allocation. Worker tenure also slightly declines (Figure 9), as firms become more cautious in sustaining marginal matches under higher separation costs. Moreover, unemployment rises across all noise regimes (Figure 10), since firms anticipate higher firing expenses and therefore become more selective in forming matches, reducing hiring and increasing unemployment. These results further validate our environment, as the observed patterns align with basic economic intuitions (Dai et al., 2015).

5.4. H4: different prior distribution of true ability

To examine how screening incentives depend on worker heterogeneity, we vary the dispersion of true ability while holding all other primitives fixed. We consider three regimes. In the baseline, worker ability is drawn from a standard normal distribution,

$$\sigma_j \sim \mathcal{N}(0,1), \qquad \hat{\sigma}_j = \sigma_j + \epsilon_j,$$

where $\hat{\sigma}_j$ is the firm's initial noisy estimate. Screening noise is

$$\epsilon_j \sim \mathcal{N}(0, \sigma_\epsilon^2), \qquad \sigma_\epsilon^2 = 2,$$

and each draw is truncated to the interval $[-2, 0.5]$ before forming $\hat{\sigma}_j$.

We then increase the dispersion of true ability to study how screening incentives respond to greater worker heterogeneity:

$$\sigma_j \sim \mathcal{N}(0,50) \quad \text{and} \quad \sigma_j \sim \mathcal{N}(0,100).$$

with the same screening noise. All other primitives are held fixed across settings.

Our results show that the *wide* regime—where worker ability is highly dispersed produces the weakest market performance. Workers exhibit the lowest average tenure (Figure 12) and persistently higher unemployment (Figure 13). Firms incur higher interview expenditures as the ability distribution widens from $\mathcal{N}(0,1)$ to $\mathcal{N}(0,50)$ and $\mathcal{N}(0,100)$ (Figure 11). Greater heterogeneity makes it more difficult to identify suitable candidates, increasing screening costs and leading to less stable, less efficient matching outcomes. Intuitively, this resembles large-scale hiring in highly heterogeneous applicant pools (e.g., Meta software engineer roles), where candidates differ substantially in productivity and interviews are both costly and necessary to avoid severe mismatches.

In contrast, in narrower and more homogeneous labor markets (e.g., standardized service jobs), worker quality is less

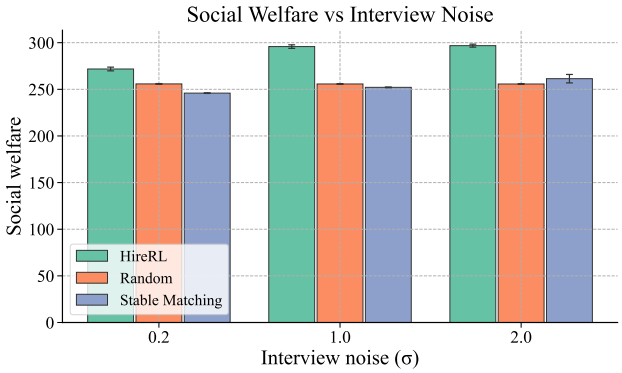

Figure 5. Social welfare under varying levels of interview noise for HireRL, a random matching baseline, and classical stable matching. Social welfare is defined as the company's reward plus workers' wages. Unemployed worker receives 0 reward. HireRL consistently achieves the highest aggregate welfare across all noise regimes, demonstrating that learned decentralized policies can better adapt to imperfect and noisy screening signals.

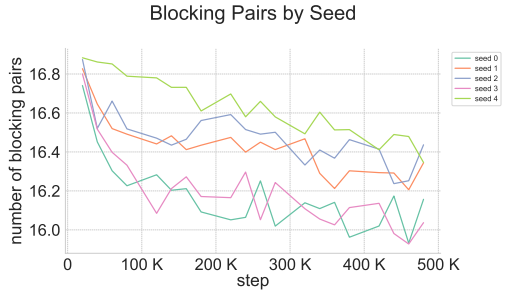

Figure 6. Average number of blocking pairs over training episodes (three random seeds). Blocking pairs are counted only when a firm makes an offer to a mutually preferred worker (offer-feasible blocking). Stable matching has zero blocking pairs by definition. Blocking pairs persist throughout training in HireRL, indicating that decentralized learning does not fully eliminate matching instability.

dispersed, and additional screening provides relatively limited informational value.

6. Discussion and Conclusion

Our results show that multi-agent learning produces economically plausible labor-market patterns: turnover declines with tenure, screening is a meaningful expenditure margin, and employment relationships remain persistent across a range of configurations. This supports HireRL as a credible testbed for studying matching with informational frictions.

Beyond validation, HireRL highlights key mechanism tradeoffs. Relative to random hiring and one-shot stable matching, learned policies improve social welfare but may increase matching instability as measured by blocking pairs. These advantages strengthen as interview signals become noisier: static mechanisms fail to adapt, whereas learning

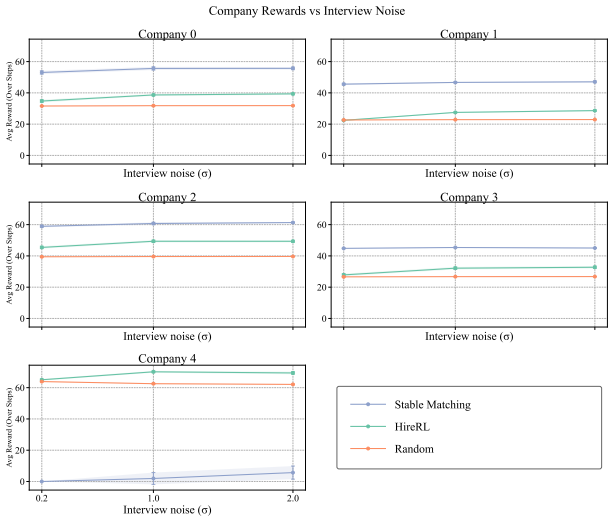

Figure 7. Reward is the firm's profit. Firm rewards under varying interview noise levels. HireRL consistently outperforms the random baseline, with performance gains becoming more pronounced as interview noise increases. Notably, HireRL achieves higher rewards only for Firm 4, which is expected in a competitive market setting where firms operate without a centralized allocation mechanism.

agents endogenously adjust screening and hiring behavior. Higher firing costs also shift outcomes toward shorter tenures, higher unemployment rates, and lower social welfare.

Methodologically, our results suggest MARL is a powerful tool for mechanism design in realistic, dynamic markets, enabling quantitative study of strategic behavior under partial observability. HireRL provides a benchmark for exploring alternative institutions, including adaptive wage rules and fairness constraints.

7. Limitations

The analogy in H4 should be interpreted with care. Our environment relies on classical economic models to map worker ability to productivity, wages, and firm profit, and does not capture the full richness of firm-specific technologies or business models. As such, the results should be understood as mechanism-level evidence of how heterogeneity and information quality reshape hiring outcomes, rather than as direct predictions for any particular employer. In this sense, our framework follows the spirit that "all models are wrong, but some are useful" (Box, 1976).

At the same time, the benchmark is naturally extensible. With access to firm-specific data—for example, calibrated profit functions, wage policies, and applicant distributions for jobs such as Meta research scientists or Uber drivers—

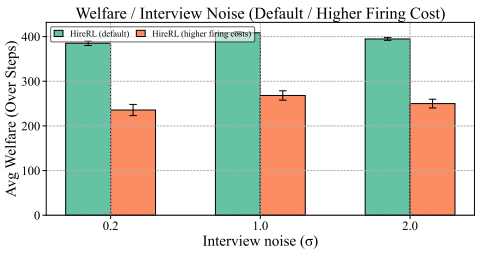

Figure 8. Average social welfare over training steps under different interview noise levels, comparing the default HireRL setting to a higher firing-cost regime. Higher firing costs consistently reduce welfare across all noise levels, while the relative performance remains stable as signal noise increases.

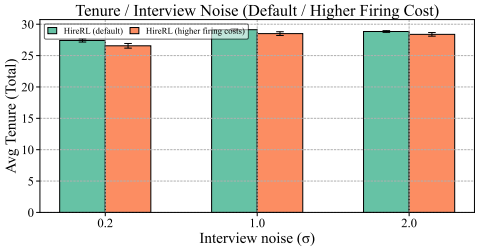

Figure 9. Average worker tenure under different interview noise levels, comparing the default HireRL setting to a higher firing-cost regime. Higher firing costs slightly reduce tenure across noise levels, as firms become more cautious in sustaining marginal matches and may adjust hiring and retention decisions in response to increased separation frictions, leading to less stable employment relationships.

HireRL could serve as a decision-support tool to study and optimize screening and hiring strategies tailored to particular labor markets. This opens the possibility of using the environment not only for theoretical comparisons but also for designing more efficient and context-specific hiring solutions for real firms.

Impact Statement

HireRL can serve as a simulation testbed for labor-market policy design, allowing regulators to explore the downstream effects of interventions (e.g., severance requirements) before deployment. By explicitly modeling costly information, it also provides insight into frictional unemployment and interventions that reduce search costs for firms and workers. HireRL has the potential to promote a more efficient and equitable labor market and support sustainable economic growth in line with the United Nations Sustainable Development Goals.

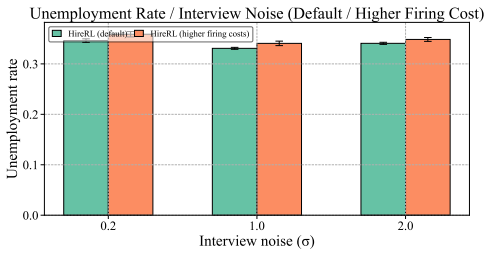

Figure 10. Unemployment rate under different interview noise levels, comparing the default HireRL setting to a higher firing-cost regime. Higher firing costs lead to consistently higher unemployment across all noise levels, since firms anticipate greater separation expenses and become more selective in forming matches, reducing overall hiring and increasing unemployment.

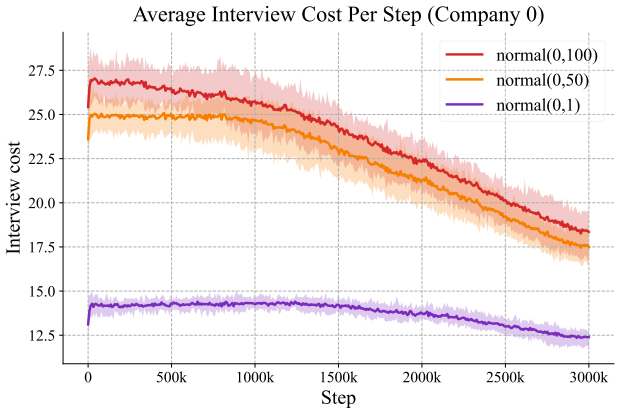

Figure 11. Average interview cost per step during training for Firm 0 under different ability distributions with the same noise level. Interview costs decline over learning, while a wider true ability distribution leads to consistently higher screening expenditures due to greater applicant heterogeneity. This pattern is consistent across all firms.

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A. Appendix: Proofs

$\ell C_{\textbf{interview}} <= 2.7$ The update for the signal is

$$\tilde{\sigma}_{j,t} = \hat{\sigma}_{j,t} - \left(\frac{(\hat{\sigma}_{j,t} - \sigma_j)^2}{1 + (\hat{\sigma}_{j,t} - \sigma_j)^2} \cdot \frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}} \right) \operatorname{sgn}\big(\hat{\sigma}_{j,t} - \sigma_j\big), \tag{5}$$

where $\ell \in (0, 1]$ is the learning rate.

We want $\hat{\sigma}_{j,t}$ and $\tilde{\sigma}_{j,t}$ to stay on the same side of the true type σ_j, i.e.

$$\operatorname{sgn}\big(\hat{\sigma}_{j,t} - \sigma_j\big) = \operatorname{sgn}\big(\tilde{\sigma}_{j,t} - \sigma_j\big).$$

This is guaranteed if the step size is never larger than the current error:

$$\left(\frac{(\hat{\sigma}_{j,t} - \sigma_j)^2}{1 + (\hat{\sigma}_{j,t} - \sigma_j)^2} \cdot \frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}} \right) \leq \big|\hat{\sigma}_{j,t} - \sigma_j\big| \quad \text{for all } \hat{\sigma}_{j,t}. \tag{6}$$

For any real number x, the function

$$\frac{|x|}{1 + x^2}$$

is maximized at $|x| = 1$ with maximum value $1/2$. Applying this with $x = \hat{\sigma}_{j,t} - \sigma_j$, we obtain

$$\frac{(\hat{\sigma}_{j,t} - \sigma_j)^2}{1 + (\hat{\sigma}_{j,t} - \sigma_j)^2} \leq \frac{1}{2} \big|\hat{\sigma}_{j,t} - \sigma_j\big|.$$

Hence

$$\text{step size} \;\leq\; \big|\hat{\sigma}_{j,t} - \sigma_j\big| \cdot \frac{1}{2} \cdot \frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}}.$$

To have the step size $\leq \big|\hat{\sigma}_{j,t} - \sigma_j\big|$ for all $\hat{\sigma}_{j,t}$, it is therefore sufficient that

$$\frac{1}{2} \cdot \frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}} \leq 1,$$

which is equivalent to

$$\frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}} \leq 2.$$

Solving this inequality gives

$$\ell C_{\text{interview}} \leq 1 + \sqrt{3} \approx 2.7.$$

Thus imposing the condition $\ell C_{\text{interview}} \leq 2.7$ guarantees that the update moves $\hat{\sigma}_{j,t}$ towards σ_j without crossing it, so $\hat{\sigma}_{j,t}$ and $\tilde{\sigma}_{j,t}$ always lie on the same side of the true type σ_j.

B. Extra Results and Graphs

With Higher dispersion on the true ability, workers exhibit the lowest average tenure (Figure 12) and persistently higher unemployment (Figure 13).

Empirical work suggests that screening is materially costly in practice (Carrillo-Tudela et al., 2023; Blatter et al., 2012). Figure 14 shows that interview expenditures constitute a non-trivial share of episode returns under decentralized learning, matching the qualitative implication that screening costs are an economically meaningful margin.

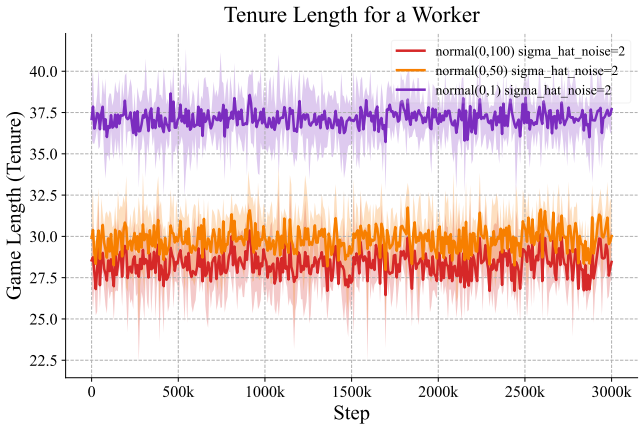

Figure 12. Average worker tenure over training episodes in the HireRL labor market under different true ability distributions. The $\mathcal{N}(0, 100)$ regime exhibits the lowest tenure, as expected: greater dispersion in worker ability makes reliable matching more difficult, resulting in shorter employment durations.

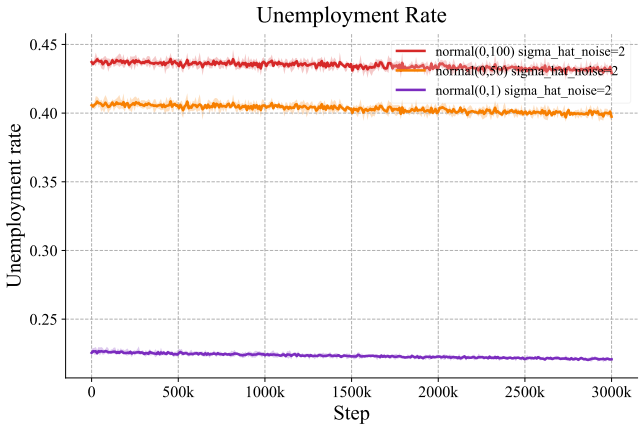

Figure 13. Unemployment rate over training steps in the HireRL labor market under different true ability distributions with the same noise level ($\hat{\sigma} = 2$). Wider ability dispersion (e.g., $\mathcal{N}(0, 100)$) leads to persistently higher unemployment, reflecting greater heterogeneity and more difficult matching, while the narrow regime ($\mathcal{N}(0, 1)$) achieves substantially lower unemployment throughout training.

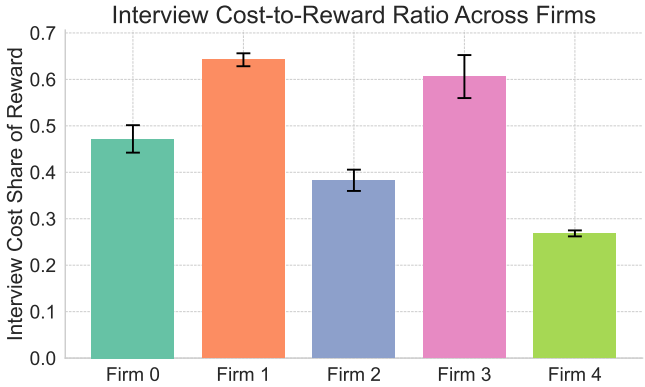

Figure 14. Interview cost share by firm. The interview cost share is defined as a firm's average interview cost per episode divided by its average episode reward, measuring the fraction of returns spent on screening. Non-negligible shares indicate that screening is an economically important decision.

C. Technical Details of the HireRL Environment

We formalize the job market as a Markov matching market and propose HireRL. HireRL comprises two sets of participants: a set of companies $\mathcal{C}$, which act as the strategic agents, and a pool of workers $\mathcal{W}$, which is modeled as part of the environment. Throughout the analysis, we assume that all agents are risk-neutral.

Reinforcement learning formulation. At each step t, the market is characterized by a global state S_t (e.g., firms' rosters, worker experience, and aggregate conditions). Each firm $i \in \mathcal{C}$ observes only $O_t^i = O(S_t)$, reflecting local information such as its workforce and noisy signals about available workers. A central decision is screening intensity: an interview investment $c_t^i \in \mathbb{R}_+$. Higher screening yields more informative signals at higher cost. Each firm follows a policy $\pi_i : O_t^i \mapsto c_t^i$ mapping observations to screening decisions.

C.1. Interview

We first model an open job board where all workers $j \in \mathcal{W}$, whether employed or unemployed, are listed. The set of workers and their attributes are treated as exogenous *environment variables* in the model (the population is fixed, and an individual worker can only transit between employment and unemployment).

Workers are born with some true ability σ_j that is drawn from a normal distribution with mean 0.5 and variance 1, truncated to the unit interval, i.e.

$$\sigma_j \sim \mathcal{N}(0.5, 1) \mid \sigma_j \in [0, 1].$$

All firms start by observing an initial PUBLIC SIGNAL OF ABILITY for each worker j, denoted $\hat{\sigma}_j \in \mathbb{R}^d$ (e.g., résumé, tests).

- Initialization: $\hat{\sigma}_{j,t} = \sigma_j + \mathcal{N}(0, \text{var}(\hat{\sigma_j}))$ with the variance term bounded between (-2, 0.5).
- At time t, if a firm decides to invest an interview cost $C_{\text{interview}}$ on worker j, then this firm observes a private signal $\tilde{\sigma}_{j,t}$. The public belief $\hat{\sigma}_{j,t}$ and the private $\tilde{\sigma}_{j,t}$ should always start from the same initial value, and both must remain on the same side of the true ability σ_j at all times. Intuitively, we want the privately updated signal $\tilde{\sigma}_j$ to contain *less noise* once a firm invests an interview cost in learning about worker j. A higher interview cost should also induce a faster convergence of the private signal toward the true type. To guarantee these properties, we specify the private update as:

$$\tilde{\sigma}_{j,t} = \hat{\sigma}_{j,t} - \underbrace{\left(\frac{(\hat{\sigma}_{j,t} - \sigma_j)^2}{1 + (\hat{\sigma}_{j,t} - \sigma_j)^2} \cdot \frac{(\ell C_{\text{interview}})^2}{1 + \ell C_{\text{interview}}} \right)}_{\text{step size}} \operatorname{sgn}\big(\hat{\sigma}_{j,t} - \sigma_j\big) \tag{7}$$

  where $\ell \in (0, 1]$ is a learning-rate parameter decides how fast should $\tilde{\sigma}_j$ converge to the true ability σ. We require $\ell C_{\text{interview}} \leq 2.7$. A proof is in the appendix A.

C.2. Matching

Perfect greedy In one timestep, firms can interview multiple workers, and workers can interview multiple firms. We apply the greedy matching algorithm, firms make offers first, and then, after workers see how much they can get from each company, pick the highest value they can get.

Initial wage offer. Each firm bases its initial wage offer on the private signal $\tilde{\sigma}_j$.

The signal is first mapped into a bounded wage component through

$$W(\tilde{\sigma}_j) = \tfrac{1}{2}\big(1 + \tanh(0.5\,\tilde{\sigma}_j)\big),$$

Among the interviewed workers, the firm selects the top $\lfloor \kappa N_{\text{int}} \rfloor$ workers—where κ is the offer rate and N_{int} is the number of interviewed workers—to receive an offer.

$$W_j = m_f\, g(\tilde{\sigma}_j)\, I_j^{\text{offer}}.$$

Workers not selected for an offer simply have $W_j = 0$.

C.3. Generate profit

A firm's primary goal is to generate profit by employing workers. We assume companies can make production with no physical capital, where a worker's human capital is the only input. Each firm (company agent) independently learns a policy (a decision-making strategy) to optimize its objective, without any central coordinator.

To formalize the firm's decision-making process, we define the state, action space, observation, and reward function for each company agent i.

- STATE (S_i): The true state of a company includes its internal properties and the latent market conditions. The state $s_{i,t} \in S_i$ is a tuple:

$$s_{i,t} = (p_{i,t}(\sigma, \exp_j), E_{i,t}, E_{o,t})$$

  where: • $p_{i,t}(\sigma) \in \mathbb{R}$ is the firm's current profit • $E_{i,t} \subseteq \mathcal{W}$ is the set of workers employed by company i at time t (as defined above). This captures the firm's immediate workforce. • $E_{o,t} \subseteq \mathcal{W}$ represents the change in the worker pool – specifically, the set of workers whose employment status changed in the last period (e.g. workers who were newly fired or hired outside of firm i).

- ACTION (A_i): The action space for a company consists of management decisions. The action $a_{i,t} \in A_i$ can be one of the following:

$$\begin{aligned} a_{i,t} \in \{\text{no-op}\} &\cup \{\text{fire}(j) \mid j \in E_{i,t}\} \\ &\cup \{\text{offer}(j, w) \mid j \in \mathcal{W} \setminus E_{i,t}, w \in \mathbb{R}^+\} \\ &\cup \{\text{interview}(j, c) \mid j \in \mathcal{W} \setminus E_{i,t}, w \in \mathbb{R}^+\} \end{aligned}$$

  where w is the proposed wage. [1]

Partial Observability (O). Each form can observe the following:

- $E_{o,t}$: current pool of available workers.

- On-the-job experience $\exp_{j,t} \in \mathbb{R}_+$ evolves deterministically while employed:

$$\begin{aligned} &\exp_{j,t+1} \\ &= \exp_{j,t} + (g_0 + g_1\sigma_j)\,\mathbf{1}\{j\text{employed at } t\}e^{-\theta \exp_{j,t}}, \\ &\text{with} \;\; \exp_{j,t} > 0, \quad \theta > 0. \end{aligned} \tag{8}$$

  Such a formulation is consistent with (Altonji & Pierret, 2001; Schönberg, 2007) where $g(\cdot)$ captures learning-by-doing and increases in ability, but exhibits diminishing marginal returns over time. High-ability workers start with a better match quality and accumulate effective experience more quickly.

- $p(\sigma, \exp_{j,t})$: realized profit, a function of true ability(unobservable) and on-the-job experience at each period. Following (Lange, 2007), when a firm i hires worker j, the match produces realized profits

$$p_{ij,t} \quad = \quad f\big(\exp_{j,t-1} + (g_0 + g_1\sigma_j)\mathbf{1}\{j \text{ employed at } t-1\}e^{-\theta \exp_{j,t-1}}\big) \;+\; \epsilon_{ij,t}, \epsilon_{ij,t} \quad \sim \quad \mathcal{N}(0, \delta_\epsilon^2). \tag{9}$$

  where $f(\cdot)$ is an increasing and diminishing–returns mapping that keeps profits on a scale comparable to worker ability. This ensures that $\mathbb{E}[p_{ij,t} \mid \sigma_j]$ is monotonic to σ_j.

[1] we do not allow a firm to directly poach employees from a competitor in the same time step – an offer can only be made to workers in the unemployed pool $\mathcal{E}t$.

- $\tilde{\sigma}_{ij,t}$: private belief about the worker's ability. Firms perfectly observe their own $p_{ij,t}$ and the experience state $\exp_{j,t-1}$, but never observe σ_j directly. Over time, the firm's posterior belief about worker quality is updated as a stable convex combination of the interview signal and the normalized profit signal:

$$\tilde{\sigma}_{ij,t+1} = (1 - v_{j,t})\,\tilde{\sigma}_{ij,\text{interview}} \;+\; v_{j,t}\,\sigma_j,$$
$$v_{j,t} = \frac{\exp_{j,t} K_1}{1 + (\exp_{j,t} - 1)K_1}, \quad K_1 = \frac{\delta^2_{\text{interview}}}{\delta^2_{\text{interview}} + \delta^2_{\varepsilon}} \tag{10}$$

  Notice that σ_j is being plugged in directly here for simplicity. Intuitively, it means, once the worker starts to generate profit, the firm can infer their true ability and update $\tilde{\sigma}_j$ accordingly. Mathematically, this formula guarantees that $v_x \in (0, 1)$, and $\tilde{\sigma}_j$ converges to σ_j.

- $\hat{\sigma}_{j,t}$: the public ability (e.g., from a resume or LinkedIn) of a worker j in any timesteps. If worker j is not hired in period t, the public belief remains unchanged.

  When the worker is hired, the public belief updates in the same direction as the private belief, but with a smaller adjustment: $\hat{\sigma}_{j,t+1} = \tilde{\sigma}_{j,t+1} + \frac{1}{2}\left[\tilde{\sigma}_{j,t+1} - \tilde{\sigma}_{j,\text{interview}}\right]$.

C.4. Wage update

Consistent with (Altonji & Pierret, 2001; Mincer, 1974; Farber & Gibbons, 1996; Schönberg, 2007), over time, wages should be more leaning toward true productivity (as measured by $\exp_j$. Here, ψ reflects the portion of match profits that the firm pays out as wages. In the quantitative implementation, we also allow for firm-size-specific wage premia $\phi_{\text{type}(i)}$ to capture the firm heterogeneity, so that the wage actually paid is

$$\begin{aligned} w_{j,t} &= \phi_{\text{type}(i)}\Big[(1 - v_x)\, g(\tilde{\sigma}_{i,j,t=\text{interview}}) + v_x \psi\, p_{ij,t-1}\Big] \\ &= \phi_{\text{type}(i)}(1 - v_x)\, g(\tilde{\sigma}_{i,j,t=\text{interview}}) \\ &\quad + \phi_{\text{type}(i)} v_x \psi\, f\Big[\exp_{j,t-2} + (g_0 + g_1\sigma_j) \\ &\quad \mathbf{1}\{j \text{ employed at } t-2\} e^{-\theta \exp_{j,t-2}} + \epsilon_{ij,t}\Big] \\ &\quad \text{where } \epsilon_{ij,t} \sim \mathcal{N}(0, \delta^2_{\epsilon}), \quad \psi \in (0,1), \quad \theta > 0. \end{aligned} \tag{11}$$

ψ reflects the portion of the profit that the company spent on wages. As t grows, $v_x \psi p_{ij,t}$ dominates. This matches the economic finding that "as experience accumulates, wages become more strongly related to variables that are likely correlated with productivity (hard-to-observe ability)."

C.5. Firing

A worker is fired if the profit they generate is too low relative to the wage the firm pays them. Specifically, the firm computes the worker's margin (profit minus wage). If this margin falls below 90% of the wage, the worker is fired. Equivalently, the worker is fired whenever:

profit $-$ wage $< 0.9 \cdot$ wage.

The reason for setting it is for numerical simplicity, which can allow us to generate some firing behaviors.

C.6. Reward

The firm's objective is to maximize the discounted sum of its future profits. The single-period reward $r_{i,t}$ for company i at time t is its net profit, calculated as the total profit generated by its workforce. The reward is formally defined as:

$$r_{i,t} = \sum_{j \in E_{i,t}} (p_{ij,t} - w_{ij,t} - c^{\text{fire}}_{i,t} - c^{\text{interview}}_{i,t}) \tag{12}$$

where $E_{i,t}$ is the set of workers employed by firm i, $p_{ij,t}$ is the match-specific profit from worker j, $w_{ij,t}$ is the wage, and $c_{i,t}^{\text{fire}}$ is the firing cost, and $c_{i,t}^{\text{hire}}$ is the hiring cost.

C.7. Timing of the game

To simply summarize what happened, we give a timeline of the game:

1. SCREENING INVESTMENT. Firms observe the public signals $\hat{\sigma}_{j,t}$ of the available workforce. For each worker j, firm i determines an interview cost $c_{ij,t} \geq 0$. This cost controls the precision of the private signal.
2. OFFER AND ACCEPTANCE. Based on the post-interview belief $\tilde{\sigma}_{ij,t=\text{interview}}$, firms extend offers to top 30% the candidates. Since the worker has not yet generated profit at the moment of hiring, the initial wage offer is determined solely by the interview signal.
3. PRODUCTION AND BELIEF UPDATE. Upon employment, the match generates a realized profit $p_{ij,t}$. The firm perfectly observes this profit and the worker's experience state, but the true ability σ_j remains latent. The firm updates its belief $\tilde{\sigma}_{ij,t+1}$ as in (10). Other firm also update their belief on public ability $\hat{\sigma}_j$
4. WAGE ADJUSTMENT. Wages are dynamically adjusted to reflect the updated belief as in (11).
5. SEPARATION (FIRING). Firms evaluate current employees using a deterministic retention rule. Notice that workers will never quit in this environment.

Note on Agency. In this formulation, the firm's strategic agency is concentrated in Step 1 (the decision of how much to invest in screening, $c_{\text{interview}}$). The subsequent steps (offering to the best candidate, updating beliefs, setting wages, and firing based on loss thresholds) are currently modeled as deterministic rules or established market mechanisms rather than learnable policy gradients.

C.8. State Transition

The evolution of the market from a global state s_t to s_{t+1} is governed by a deterministic transition function T, conditioned on the joint action A_t taken by all agents. The global state is the aggregation of all individual agent states: $s_t = (\{s_{i,t}\}_{i \in \mathcal{C}})$. The joint action $A_t = (\{a_{i,t}\}_{i \in \mathcal{C}})$ encapsulates all decisions made by companies at time t.

The transition $s_{t+1} = T(s_t, A_t)$ can be broken down into two logical phases: ACTION RESOLUTION and STATE EVOLUTION.

C.8.1. ACTION RESOLUTION

In this phase, the environment processes the joint action A_t to determine the immediate outcomes. Unlike models with an external stochastic matching mechanism, here the matching is a direct and deterministic result of the agents' explicit `offer`, `accept`, and `fire` actions.

We resolve the actions into two key sets:

- New Hires ($\mathcal{H}_t$): A set of new company-worker pairs. A pair (i, j) is added to $\mathcal{H}_t$ if company i executes offer(j, w) and worker j executes accept(i).

$$\mathcal{H}_t = \{(i, j) \mid a_{i,t} = \text{offer}(j, w) \wedge a_{j,t} = \text{accept}(i)\}$$

- Separations ($\mathcal{F}_t$): A set of company-worker pairs that are dissolved. A pair (i, j) is added to $\mathcal{F}_t$ if company i executes fire(j).

$$\mathcal{F}_t = \{(i, j) \mid a_{i,t} = \text{fire}(j)\}$$

Given that all actions are simultaneous, we make an implicit assumption on timing: firings and hirings in the same period are resolved such that a worker fired at time t is considered unemployed in t (eligible to be hired by someone else at the next time step $t + 1$, but not in the same instant t). Likewise, a worker hired at time t becomes part of their new firm's workforce from that point onward (effective for production in the next period). Because we disallow offers to currently employed workers, we avoid any direct conflict between $\mathcal{H}_t$ and $\mathcal{F}_t$ (a worker cannot be fired and hired in the same period by two different firms).

C.8.2. State Evolution

After resolving actions, the environment updates both company and worker states:

- Workforce update:
$$E_{i,t+1} = (E_{i,t} \setminus \{j \mid (i,j) \in \mathcal{F}_t\}) \cup \{j \mid (i,j) \in \mathcal{H}_t\}.$$
- Experience, private and public ability transition: Experience as in equation 8, private ability as in equation 7 and public ability as in C.3